Representability of partial sums in generic absolute and positive web models

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Allowed standard prerequisites (editorial).

- Symmetric monoidal closed categories, currying naturality, Seely categorical notation and basic partial-sum axioms. Original references22 Hines2013 doi10.1017/S0960129512000059 and38 Mellies2009, with exact current definitions and current author matrix/predual proofs retained. Proposition33 original proof pointer is retained and points to collected Lemma21/Theorem23 proofs, not an invented standard-prerequisite exemption.

Partial commutative sums: original Definitions 1–4, 10 and 12

If A is a countable set and M is a set, an A-indexed family of elements of M is a function $\vec{x} : A \to M$, written $\vec{x} = (x_a)_{a\in A}$. We consider a set M with a partial function Σ called the sum from indexed families on M to M. An indexed family $(x_a)_{a\in A}$ is summable if it is in the domain of Σ, and we write its image as $\sum_{a\in A} x_a$.

Original definition (1)**.** *For any expression e and e' involving partial sums, we write $e \longrightarrow e'$ (or $e' \longleftarrow e$) if whenever e is defined, then e' is also defined and $e = e'$. We write $e \multimap\!\!\!\shortmid e'$ if $e \longrightarrow e'$ and $e' \longrightarrow e$.*

Original definition (2, original reference 22)**.** *The tuple (M, Σ) is a partial commutative monoid (PCM) if M is non-empty and if the sum Σ satisfies the following axioms.*

(Unary) *Unary sum axiom: every singleton family (x) is summable with sum x.*

(WPA) *Weak partition associativity axiom: Let $(x_a)_{a\in A}$ be an indexed family, I countable, and $\{A_i\}_{i\in I}$ a partition of A (the A_i's can be empty). Then*

$$\sum_{a\in A} x_a \longrightarrow \sum_{i\in I} \left(\sum_{a\in A_i} x_a \right).$$

A PCM is strong if it satisfies the partition associativity axiom (PA), where the implication $\longrightarrow$ of (WPA) is replaced by an equivalence $\multimap\!\!\!\shortmid$.

Observe that the neutral element 0 is not given as part of the data, but can be defined from the axioms. Indeed, the sum over the empty family is always defined by Lemma 3.

Original lemma (3)**.** *If $A \subseteq B$ and $(x_b)_{b\in B}$ is summable, then $(x_a)_{a\in A}$ is summable.*

Original definition (4)**.** *Every PCM has a canonical preorder (it is not antisymmetric in general) given by $x \leq y$ if there exists z such that $x + z = y$.*

Original definition (10). *A Partial Commutative Rig (PCR) is the data of $(R, \Sigma, 1, \cdot)$ such that (R, Σ) is a PCM, $(R, 1, \cdot)$ is a commutative monoid, and such that for all indexed families $(x_a)_{a\in A}$ and $(y_b)_{b\in B}$ of R,*

$$\left(\sum_{a\in A} x_a\right) \cdot \left(\sum_{b\in B} y_b\right) \longrightarrow \sum_{a\in A, b\in B} (x_a \cdot y_b).$$

A PCR is strong if the underlying PCM is strong.

Original definition (12, original reference 22). *Let L be a locally small category such that for all objects X, Y, $L(X, Y)$ is a PCM. The category L is a PCM category if for all $(f_a \in L(X, Y))_{a\in A}$ and $(g_b \in L(Y, Z))_{b\in B}$,*

$$\left(\sum_{b\in B} g_b\right) \circ \left(\sum_{a\in A} f_a\right) \longrightarrow \sum_{(a,b)\in A\times B} (g_b \circ f_a).$$

Web models and predual characterization: original Part I

We define the Kronecker symbol: $\delta_{ab} = 0$ if $a \neq b$, and $\delta_{aa} = 1$. We define for all $a \in |X|$ a vector $e_a \in R^{|X|}$ by $(e_a)_{a'} = \delta_{aa'}$. For all set $|X|$ (called the web) and all PCR R, we define the following partial maps on vectors:

$$\langle x \mid y\rangle_R = \sum_{a\in|X|} x_a y_a, \qquad (t \cdot s)_{a,c} = \sum_{b\in|Y|} s_{a,b} \cdot t_{b,c}, \qquad (s \cdot x)_b = \sum_{a\in|X|} s_{a,b} x_a.$$

Let L be a model of linear logic, more precisely a Seely category. We write the composition of $f \in L(X, Y)$ with $g \in L(Y, Z)$ as $g \circ f$. We write the symmetric monoidal structure as $(\otimes, \lambda, \rho, \alpha, \gamma)$ where $\lambda_X : 1 \otimes X \to X$ and $\rho_X : X \otimes 1 \to X$ are the unitors, $\alpha_{X,Y,Z} : (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z)$ is the associator, and $\gamma_{X,Y} : X \otimes Y \to Y \otimes X$ is the symmetry. We write $X \multimap Y$ the internal hom, the currying of $f : X \otimes Y \to Z$ is $\mathrm{cur}(f) : X \to (Y \multimap Z)$, and the evaluation is $\mathrm{ev} : (X \multimap Y) \otimes X \to Y$. First we assume that R_+ is a strong PCR, and we consider $B \subseteq R_+$. We assume that B is downward closed: if $x \leq y$ and $y \in B$ then $x \in B$. We define an orthogonality relation $x \perp y$ on vectors and the orthogonal of a set $F \subseteq R_+^{|X|}$ as:

$$x \perp y \text{ if } \langle x \mid y\rangle_R \text{ is defined and belongs to } B, \qquad F^\perp = \{x' \in R_+^{|X|} \mid x \perp x'\}.$$

This construction has the usual properties of an orthogonality construction: $F \subseteq F^{\perp\perp}$, $(F \subseteq G) \Rightarrow (G^\perp \subseteq F^\perp)$, and $F^{\perp\perp\perp} = F^\perp$.

Original definition (14). *A set of vectors $F \subseteq R_+^{|X|}$ is a covering if for all $a \in |X|$, there exists $x \in F$ such that its a component x_a is invertible.*

Original definition (15). *An R_+-space is a pair $X = (|X|, AX)$ where $|X|$ is a set and $AX \subseteq R_+^{|X|}$ is a set of vectors such that $AX^{\perp\perp} = AX$, and such that both AX and $AX^\perp$ are coverings. The dual of an R_+-space X is defined as the R_+-space $X^\perp = (|X|, AX^\perp)$.*

Original lemma (16, Downward closure). *For all $x, y \in R_+^{|X|}$ such that $x \leq y$, $y \in AX \Rightarrow x \in AX$.*

For all vectors $x \in R_+^{|X|}$ and $y \in R_+^{|Y|}$, define $x \otimes y \in R_+^{|X|\times|Y|}$ by $(x \otimes y)_{a,b} = x_a y_b$. For all sets $F \subseteq R_+^{|X|}$ and $G \subseteq R_+^{|Y|}$, we set $F \otimes G = \{x \otimes y \mid x \in F, y \in G\}$. The tensor $X \otimes Y$ of two R_+-spaces X and Y is defined as

$$|X \otimes Y| = |X| \times |Y|, \qquad A(X \otimes Y) = (AX \otimes AY)^{\perp\perp}.$$

We now define the R_+-space of linear maps between R_+-spaces by duality: $X \multimap Y = (X \otimes Y^\perp)^\perp$.

Remark 18. $A(X \multimap Y) = A(X \otimes Y^\perp)^\perp = (AX \otimes AY^\perp)^{\perp\perp\perp} = (AX \otimes AY^\perp)^\perp$. Thus, a matrix $s \in R^{|X\multimap Y|}$ belongs to $A(X \multimap Y)$ if and only if $s \perp (x \otimes y')$ for all $x \in AX$, $y' \in AY^\perp$.

Original theorem (19). *For all $s \in A(X \multimap Y)$ and $t \in A(Y \multimap Z)$, $t \cdot s$ is well-defined and belongs to $A(X \multimap Z)$. Furthermore, the identity matrix $\mathrm{id}_X \in R_+^{|X\multimap X|}$ belongs to $A(X \multimap X)$.*

Thus, we can define the category Web_{R_+} whose objects are the R_+-spaces, whose morphisms are $\mathsf{Web}_{R_+}(X,Y) = A(X \multimap Y)$, and whose composition is given by matrix multiplication.

Original lemma (20). *For all $s \in R_+^{|X|\times|Y|}$, $x \in R_+^{|X|}$ and $y \in R_+^{|Y|}$,*

$$\langle s \cdot x \mid y \rangle \longrightarrow \langle s \mid x \otimes y \rangle.$$

Furthermore, if $s \cdot x$ is defined, then $\langle s \cdot x \mid y \rangle \multimap\!\!\!\shortrightarrow \langle s \mid x \otimes y \rangle$.

$$\langle s^\perp \cdot y \mid x \rangle \longrightarrow \langle s \mid x \otimes y \rangle.$$

Furthermore, if $s^\perp \cdot y$ is defined, then $\langle s^\perp \cdot y \mid x \rangle \multimap\!\!\!\shortrightarrow \langle s \mid x \otimes y \rangle$. Where $s^\perp \in R_+^{|Y|\times|X|}$ is the transpose of s, $(s^\perp)_{b,a} = s_{a,b}$.

Original lemma (21, Covering principle). *Let Q be a covering of $R_+^{|Y|}$. Let $x \in R_+^{|X|}$ and $s \in R_+^{|X\multimap Y|}$ such that $\langle s \mid x \otimes y \rangle$ is well-defined for all $y \in Q$. Then $s \cdot x$ is well-defined. Similarly, let P be a covering of $R_+^{|X|}$. Let $y \in R_+^{|Y|}$ and $s \in R_+^{|X\multimap Y|}$ such that $\langle s \mid x \otimes y \rangle$ is well-defined for all $x \in P$. Then $s^\perp \cdot y$ is well-defined.*

Original proposition (22). *Let $s \in R_+^{|X\multimap Y|}$. The following are equivalent:*

1. *$s \in A(X \multimap Y)$;*
2. *$s^\perp \in A(Y^\perp \multimap X^\perp)$;*
3. *$\forall x \in AX$, $s \cdot x$ is defined and belongs to AY;*
4. *$\forall y' \in AY^\perp$, $s^\perp \cdot y'$ is defined and belongs to $AX^\perp$.*

Original theorem (23, Predual characterization of morphisms). *Let X and Y be two R_+-spaces. Let $P \subseteq R_+^{|X|}$ and $Q \subseteq R_+^{|Y|}$ be two coverings such that $P^{\perp\perp} = AX$ and $Q^\perp = AY$. Then for all $s \in R_+^{|X\multimap Y|}$, the following are equivalent.*

1. *$s \in A(X \multimap Y)$;*
2. *$s \in (P \otimes Q)^\perp$;*
3. *for all $x \in P$, $s \cdot x$ is defined and belongs to AY;*
4. *for all $y' \in Q$, $s^\perp \cdot y'$ is defined and belongs to $AX^\perp$.*

Original corollary (24). *Let $P \subseteq R_+^{|X|}$ and $Q \subseteq R_+^{|Y|}$ be two coverings such that $P^{\perp\perp} = AX$ and $Q^{\perp\perp} = AY$. Then $A(X \otimes Y) = (P \otimes Q)^{\perp\perp}$.*

The following result immediately follows from Theorem 23, using that $X \otimes Y = (X \multimap Y^\perp)^\perp$.

Original definition (26). *A PCR R is absolute if there exists a strong PCR R_+ and a function $|.| : R \to R_+$ (called the absolute value) such that*

- *$|x| = 0$ if and only if $x = 0$;*
- *$|.|$ is a monoid morphism for the multiplicative part: $|xy| = |x||y|$ and $|1| = 1$;*
- *Triangle inequality: $(x_a)_{a\in A}$ is summable in R if and only if $(|x_a|)_{a\in A}$ is summable in R_+, and $|\sum_{a\in A} x_a| \leq \sum_{a\in A} |x_a|$.*

The absolute value $|.|$ extends to a function $\mathrm{abs} : R^{|X|} \to R_+^{|X|}$ for all web $|X|$, defined by $\mathrm{abs}(x)_a = |x_a|$ for all $a \in |X|$.

Original definition (28). *For all R_+-space X, let $R\langle X\rangle = \{x \in R^{|X|} \mid \mathrm{abs}(x) \in AX\}$. We define the category Web_R whose objects are the R_+-spaces, whose morphisms from X to Y are the matrices in $R\langle X \multimap Y\rangle$, and whose composition is matrix multiplication.*

For all $s \in R\langle X \multimap Y\rangle$ and $t \in R\langle Y \multimap Z\rangle$, we have that $\mathrm{abs}(t \cdot s) \multimap\!\!\!\shortleftarrow \mathrm{abs}(t) \cdot \mathrm{abs}(s) \in A(X \multimap Z)$, so $t \cdot s$ is well-defined and belongs to $R\langle X \multimap Z\rangle$ by downward closure of $A(X \multimap Z)$ (Lemma 16).

Original definition (30)**.** *Let X be an R_+-space. A family $(x(i))_{i\in I}$ of elements of AX is summable if their pointwise sum $(\sum_{i\in I} x(i))_a = \sum_{i\in I} x(i)_a$ is defined and belong to AX.*

Original proposition (31)**.** *Let X be an R_+-space. Then AX is a strong PCM.*

Proof. It immediately follows from (PA) on R_+ that for all family $(x(i))_{i\in I}$ of $R_+^{|X|}$ and all partition $\{I_k\}_{k\in K}$ of I, $\sum_{i\in I} x(i) \multimap \sum_{k\in K}\sum_{i\in I_k} x(i)$. To conclude, we still need to ensure that whenever $(x(i))_{i\in I}$ is summable in AX, then for all $k \in K$, $(x(i))_{i\in I_k}$ is summable in AX. This is an immediate consequence of the downward closure of AX (Lemma 16). □

Similarly to morphisms, summability can be characterized in terms of predual. The proof is very similar to the proof of Lemma 21 and of Theorem 23.

Original proposition (33)**.** *Let P be a covering of $R_+^{|X|}$ such that $P^\perp = AX$. Then a family $(x(i))_{i\in I}$ of AX is summable if and only if for all $x' \in P$, $\langle \sum_{i\in I} x(i) \mid x' \rangle$ is defined and is in B.*

Original definition (34)**.** *Let X be an R_+-space. A family $(x(i))_{i\in I}$ of elements of $R\langle X\rangle$ is summable if $(\mathrm{abs}(x(i)))_{i\in I}$ is summable in AX (in the sense of Definition 30).*

Original proposition (35)**.** *The pointwise sum defined above is a PCM on $R\langle X\rangle$.*

Proof. Let $(x(i))_{i\in I}$ be summable in $R\langle X\rangle$, and $\{I_k\}_{k\in K}$ be a partition of I. For all $k \in K$, $(\mathrm{abs}(x(i)))_{i\in I_k}$ is summable in AX by Lemma 3. Thus, $(x(i))_{i\in I_k}$ is summable in $R\langle X\rangle$:

$$\sum_{k\in K} \mathrm{abs}\left(\sum_{i\in I_k} x(i)\right) \overset{1}{\leq} \sum_{k\in K}\sum_{i\in I_k} \mathrm{abs}(x(i)) \overset{2}{\multimap} \sum_{a\in A} \mathrm{abs}(x_a) \in AX$$

where (1) holds by triangle inequality and (2) by (PA) on the strong PCM AX. We conclude that $(\sum_{i\in I_k} x(i))_{k\in K}$ is summable in $R\langle X\rangle$ by downward closure of AX. □

Original definition (37)**.** *A summability structure on a category L is a tuple $(S, (\pi_i)_{i\in\mathbb{N}}, \sigma, 0)$ where*

(S-fun) *S is a functor on L, and $\pi_i, \sigma : S \Rightarrow \mathrm{Id}$ are natural transformations called respectively projections and sum, and 0 is a family of zero morphisms $0_{X,Y} \in L(X,Y)$.*

(S-proj) *The π_i are jointly monic: $\forall f, g \in L(X, SY)$, $(\forall i \in \mathbb{N}, \pi_i \circ f = \pi_i \circ g) \Rightarrow f = g$.*

(S-sum) *For all objects X, Y, there exists a PCM structure on $L(X,Y)$ such that*

(S-sum-1) A family $(f_{i_1,\dots,i_n})_{i_1,\dots,i_n\in\mathbb{N}}$ over $L(X,Y)$ is summable if and only if there exists $f \in L(X, S^nY)$ such that $\pi_{i_1} \circ \cdots \circ \pi_{i_n} \circ f = f_{i_1,\dots,i_n}$.

(S-sum-2) If $(f_{i_1,\dots,i_n})_{i_1,\dots,i_n\in\mathbb{N}}$ is summable then $\sum_{i_1,\dots,i_n\in\mathbb{N}} f_{i_1,\dots,i_n} = \underbrace{\sigma \circ \cdots \circ \sigma}_{n} \circ f$.

(S-sum-3) The sum over the empty family is equal to the zero morphism 0.

Representability: original Definitions 49–50 and Proposition 52

We assume in this section that L is a symmetric monoidal closed category, with zero morphisms 0 that are also absorbing for the monoidal product: $\mathrm{id}_X \otimes 0 = 0$ and $0 \otimes \mathrm{id}_Y = 0$.

Original definition (49)**.** *A summation object in L is an object D equipped with a family of jointly epic morphisms $(e_i : 1 \to D)_{i\in\mathbb{N}}$ called the basis, and $\Delta : 1 \to D$ called the diagonal.*

The internal hom $1 \multimap X$ can be identified with X, and then $\mathrm{ev}_{1,X} = \rho_X : X \otimes 1 \to X$ and for all $f : X \otimes 1 \to Y$, $\mathrm{cur}(f) = f \circ \rho_X^{-1}$. We define natural transformations $\pi_i = (e_i \multimap X) : (D \multimap X) \to X$ and $\sigma = (\Delta \multimap X) : (D \multimap X) \to X$. More explicitly,

$$\pi_i = \mathrm{ev}_{D,X} \circ ((D \multimap X) \otimes e_i) \circ \rho^{-1}, \qquad \sigma = \mathrm{ev}_{D,X} \circ ((D \multimap X) \otimes \Delta) \circ \rho^{-1}. \tag{3}$$

Original definition (50)**.** *The category L is a representable PCM category if it has a summation object D such that $(D \multimap _, (\pi_i)_{i\in\mathbb{N}}, \sigma, 0)$ is a summability structure.*

We now provide an alternative characterization of representable PCM categories, which can be obtained by "uncurrying" the definition of summability structures. By standard naturality equations on the currying, we get that for all $f : X \otimes D \to Y$ and $g : X \otimes D^{\otimes n} \to Y$,

$$\begin{aligned} \pi_i \circ \mathrm{cur}(f) &= f \circ (X \otimes e_i) \circ \rho^{-1}, & \sigma \circ \mathrm{cur}(f) &= f \circ (X \otimes \Delta) \circ \rho^{-1}, \\ \pi_{i_1} \circ \cdots \circ \pi_{i_n} \circ \mathrm{cur}^n(g) &= g \circ (X \otimes e_{i_n} \otimes \cdots \otimes e_{i_1}) \circ (\rho^n)^{-1}, \\ \sigma \circ \cdots \circ \sigma \circ \mathrm{cur}^n(g) &= g \circ (X \otimes \Delta \otimes \cdots \circ \Delta) \circ (\rho^n)^{-1}. \end{aligned} \tag{4}$$

Original lemma (51)**.** *For all object D and every family of morphisms $(e_i : 1 \to D)_{i \in \mathbb{N}}$, define π_i as in Equation (3). Then, the following assertions are equivalent:*

1. *the morphisms e_i are jointly epic;*
2. *for all object X, the morphisms $X \otimes e_i : X \otimes 1 \to X \otimes D$ are jointly epic;*
3. *the morphisms π_i are jointly monic.*

Proof. (1) $\Rightarrow$ (2). Let $f, g : X \otimes D \to Y$ be such that $f \circ (X \otimes e_i) = g \circ (X \otimes e_i)$ for all $i \in \mathbb{N}$. By naturality of γ, we get $f \circ \gamma \circ (e_i \otimes X) = g \circ \gamma \circ (e_i \otimes X)$ and hence $\mathrm{cur}(f \circ \gamma) \circ e_i = \mathrm{cur}(g \circ \gamma) \circ e_i$ so that $\mathrm{cur}(f \circ \gamma) = \mathrm{cur}(g \circ \gamma)$ and hence $f = g$. So the $X \otimes e_i$ are jointly epic. Conversely, (2) $\Rightarrow$ (1), taking $X = 1$. Finally, (2) $\Leftrightarrow$ (3) by Equation (4) and bijectivity of cur. □

Thus, the only condition that does not hold de facto when there is a summation object is (S-sum). This condition can be uncurried by Equation (4) as follows.

Original proposition (52)**.** *The category L is a representable PCM category if and only if it has a summation object D such that for all objects X, Y, $L(X, Y)$ has a PCM structure in which:*

1. *A family $(f_{i_1,\ldots,i_n})_{i_n,\ldots,i_n \in \mathbb{N}}$ of morphisms $X \to Y$ is summable if and only if there exists $h : X \otimes D^{\otimes n} \to Y$ such that $h \circ (X \otimes e_{i_n} \otimes \cdots \otimes e_{i_1}) \circ (\rho^n_X)^{-1} = h_{i_1,\ldots,i_n}$.*
2. *Then, $\sum_{i_1,\ldots,i_n \in \mathbb{N}} h_{i_1,\ldots,i_n} = h \circ (X \otimes \Delta \otimes \cdots \otimes \Delta) \circ (\rho^n_X)^{-1}$.*
3. *The sum over the empty family is equal to the zero morphism 0.*

Original Section 7: Taylor expansion in web models

We prove in this section that our web models are representable PCM categories, and feature a Taylor expansion. For all set A and all PCR R, define $\Delta_A \in R^A$ by $(\Delta_A)_a = 1$. We then define $l^1(A) \subseteq R^A$ and $l^\infty(A) \subseteq R^A$ by $l^1(A) = \{\Delta_A\}^\perp$ and $l^\infty(A) = \{\Delta_A\}^{\perp\perp}$. In particular, $(x_a)_{a \in A} \in l^1(A)$ if and only if $(x_a)_{a \in A}$ is summable and $\sum_{a \in A} x_a \in B$. Then, we set $D = (\mathbb{N}, l^\infty(\mathbb{N}))$. The object D is a summation object, with diagonal $\Delta = \Delta_{\mathbb{N}}$ and basis $e_i \in R^{|D|}$ defined by $(e_a)_{a'} = \delta_{aa'}$ (they belong to AD by downward closure, since $e_a \leq \Delta$). We now prove that the summability in Web_R described in Section 4 is representable.

Original lemma (58)**.** *For all $k \in \mathbb{N}$, $A(D^{\otimes k}) = l^\infty(\mathbb{N}^k)$.*

Proof. $A(D^{\otimes k}) = (\{\Delta\}^{\perp\perp} \otimes \cdots \otimes \{\Delta\}^{\perp\perp})^{\perp\perp} \overset{1}{=} (\{\Delta\} \otimes \cdots \otimes \{\Delta\})^{\perp\perp} = \{\Delta_{\mathbb{N}^k}\}^{\perp\perp} = l^\infty(\mathbb{N}^k)$ where (1) is obtained by iterating Corollary 24. □

Original theorem (59)**.** *The sum in Web_{R_+} and Web_R is representable.*

Proof. It suffices to do the proof for Web_R (Web_{R_+} is a special instance of Web_R in which $|.|$ is the identity function). We prove that D satisfies the conditions of Proposition 52. Let $(f(i_1, \ldots, i_k))_{i_1,\ldots,i_k \in \mathbb{N}}$ be a family of $R\langle X \multimap Y \rangle$. Define $h \in R^{|X \otimes D \multimap Y|}$ by $h_{a,i_k,\ldots,i_1,b} = f(i_1, \ldots, i_k)_{a,b}$ so that

$$f(i_1, \ldots, i_k) = h \cdot (X \otimes e_{i_k} \otimes \cdots \otimes e_{i_1}) \cdot (\rho^k)^{-1}$$

and

$$\sum_{i_1,\ldots,i_k \in \mathbb{N}} f(i_1, \ldots, i_k) \multimap\!\!\!\shortrightarrow h \cdot (X \otimes \Delta \otimes \cdots \otimes \Delta) \cdot (\rho^k)^{-1} \multimap\!\!\!\shortrightarrow h \cdot (X \otimes \Delta_{\mathbb{N}^k}) \cdot (\rho^k)^{-1}.$$

It suffices to prove that $(f(i_1,\dots,i_k))_{i_1,\dots,i_k\in\mathbb{N}}$ is summable in $R\langle X \multimap Y\rangle$ if and only if $h \in R\langle X\otimes D^{\otimes k} \multimap Y\rangle$ to conclude. We proceed by the following chain of equivalences.

$$\begin{aligned}
&(f_{i_1,\dots,i_k})_{i_1,\dots,i_k\in\mathbb{N}} \text{ is summable in } R\langle X \multimap Y\rangle\\
\iff &(\mathrm{abs}(f)_{i_1,\dots,i_k})_{i_1,\dots,i_k\in\mathbb{N}} \text{ is summable in } A(X \multimap Y)\\
\overset{1}{\iff} &\forall x \in AX, \forall y' \in AY^\perp, \langle \sum_{i_1,\dots,i_k\in\mathbb{N}} \mathrm{abs}(f)_{i_1,\dots,i_k} \mid x\otimes y'\rangle \text{ is defined and in } B\\
\overset{2}{\iff} &\forall x \in AX, \forall y' \in AY^\perp, \langle \mathrm{abs}(h) \mid x\otimes \Delta_{\mathbb{N}^k}\otimes y'\rangle \text{ is defined and in } B\\
\overset{3}{\iff} &\mathrm{abs}(h) \in A(X\otimes D^{\otimes k} \multimap Y) \iff h \in R\langle X\otimes D^{\otimes k} \multimap Y\rangle.
\end{aligned}$$

where (1) is the predual characterization of summability given in Proposition 33, (2) holds by the covering principle of Lemma 21 ($\{\Delta_{\mathbb{N}}\}$ is a covering), and (3) holds by the predual characterization of Theorem 23, using that $A(D^{\otimes k}) = \{\Delta_{\mathbb{N}^k}\}^{\perp\perp}$ by Lemma 58. □

Original Appendix A: Proofs on web based models

Proof of Lemma 20. Observe that

$$\langle s\cdot x \mid y\rangle = \sum_{b\in|Y|}\left(\sum_{a\in|X|} s_{a,b}x_a\right) y_b \xrightarrow{1} \sum_{b\in|Y|}\sum_{a\in|X|}(s_{a,b}x_a y_b) \xrightarrow{2} \sum_{a\in|X|,b\in|Y|} s_{a,b}x_a y_b = \langle s \mid x\otimes y\rangle$$

where (1) is by distributivity of multiplication over sums, and (2) is by (PA). Furthermore, if $s\cdot x$ is defined, then (1) turns into an equivalence $\multimap\!\!\!\!\!\!\rightarrow$ (but not in general, consider $y_b = 0$). The second item is proved in the same way.

Proof of Lemma 21. Let $b_0 \in |Y|$. We prove that $(s\cdot x)_{b_0}$ is defined. By covering, there exists $y \in F$ such that y_{b_0} is invertible. By assumption, $(s_{a,b}x_a y_b)_{a\in|X|,b\in|Y|}$ is summable, so by Lemma 3 $(s_{a,b_0}x_a y_{b_0})_{a\in|X|}$ is summable. But then

$$y_{b_0}^{-1}\left(\sum_{a\in|X|} s_{a,b_0}x_a y_{b_0}\right) \longrightarrow \sum_{a\in|X|} s_{a,b_0}x_a y_{b_0}y_{b_0}^{-1} = \sum_{a\in|X|} s_{a,b_0}x_a$$

so $(s_{a,b_0}x_a)_{a\in|X|}$ is summable.

Proof of Proposition 22. We first prove (1) ⇔ (2). By Remark 18,

$$\begin{aligned}
s \in A(X \multimap Y) &\iff \forall x \in AX, \forall y' \in AY, \langle s \mid x\otimes y'\rangle \text{ is defined and in } B,\\
s^\perp \in A(Y^\perp \multimap X^\perp) &\iff \forall x \in AX, \forall y' \in AY, \langle s^\perp \mid y'\otimes x\rangle \text{ is defined and in } B.
\end{aligned}$$

We can easily check that $\langle s \mid x\otimes y'\rangle \multimap\!\!\!\!\!\!\rightarrow \langle s^\perp \mid y'\otimes x\rangle$, which concludes the proof of equivalence. Next, we prove (1) ⇒ (3). Let $s \in A(X \multimap Y)$ and $x \in AX$. First, observe that $s\cdot x$ is well-defined by covering principle (Lemma 21), since $AY^\perp$ is a covering. Then, for all $y' \in Y^\perp$, $\langle s\cdot x \mid y'\rangle \multimap\!\!\!\!\!\!\rightarrow \langle s \mid x\otimes y'\rangle$ by Lemma 20. Since $s \perp x\otimes y'$ by assumption, we get that $s\cdot x \perp y'$. It concludes the proof of (3). The proof of the reverse implication (3) ⇒ (1) relies on Lemma 20 in a similar same way. Finally, (2) ⇔ (4) holds using (1) ⇔ (3) on $s^\perp$.

Proof of Theorem 19. Let $s \in A(X \multimap Y)$ and $t \in A(Y \multimap Z)$. First, let us prove that $t\cdot s$ is well-defined. By covering, for all $a_0 \in |X|$ there exists $x \in AX$ such that x_{a_0} is invertible. Then, $(t\cdot(s\cdot x))$ is defined by iterating Proposition 22 twice. But

$$(t\cdot(s\cdot x))_c = \sum_{b\in|Y|} t_{b,c}\left(\sum_{a\in|X|} s_{a,b}x_a\right) \longrightarrow \sum_{b\in|Y|}\sum_{a\in|X|} t_{b,c}s_{a,b}x_a \xrightarrow{\text{PA}} \sum_{a\in|X|,b\in|Y|} s_{a,b}t_{b,c}x_a.$$

It follows from (WPA) that $(s_{a_0,b}t_{b,x}x_{a_0})_{b\in|Y|}$ is summable, so by invertibility of x_{a_0}, $(s_{a_0,b}t_{b,c})_{b\in|Y|}$ is summable. That is, $t \cdot s$ is well-defined. We now prove that it belongs to $A(X \multimap Z)$ using Proposition 22. Let $x \in AX$. Then,

$$((t \cdot s) \cdot x)_c = \sum_{a\in|X|} \left(\sum_{b\in|Y|} s_{a,b}t_{b,c} \right) x_a \xrightarrow{1} \sum_{a\in|X|} \left(\sum_{b\in|Y|} s_{a,b}t_{b,c}x_a \right),$$

$$(t \cdot (s \cdot x))_c = \sum_{b\in|Y|} t_{b,c} \left(\sum_{a\in|X|} s_{a,b}x_a \right) \xrightarrow{2} \sum_{b\in|Y|} \left(\sum_{a\in|X|} s_{a,b}t_{b,c}x_a \right).$$

Be aware: (1) only holds because we already know that $t \cdot s$ is defined, and (2) only holds because we already know that $s \cdot x$ is defined. Then, by (PA), $(t \cdot s) \cdot x \multimap\!\!\!\!\!\!\leftrightarrow t \cdot (s \cdot x)$. We conclude that $t \cdot s$ is well-defined and belongs to $A(X \multimap Z)$ by iterating Proposition 22 twice.

Proof of Theorem 23. First, observe that (1) $\Rightarrow$ (2) is trivial.

(2) $\Rightarrow$ (3). Let $s \in (P \otimes Q)^\perp$. Then for all $x \in P$, $s \cdot x$ is well-defined by covering principle (Lemma 21). Furthermore, for all $y' \in Q$, $\langle s \mid x \otimes y' \rangle \multimap\!\!\!\!\!\!\leftrightarrow \langle s \cdot x \mid y' \rangle$ by Lemma 20. But $s \perp x \otimes y'$ so we deduce that $s \cdot x \perp y'$. Thus, $s \cdot x \in Q^\perp = AY$.

(3) $\Rightarrow$ (2). Assume that s satisfies (3). Then we prove that $s \in (P \otimes AY^\perp)^\perp \subseteq (P \otimes Q)^\perp$. Let $x \in P$ and $y' \in (AY)^\perp$. Then by assumption $s \cdot x$ is defined so by Lemma 20 $\langle s \mid x \otimes y' \rangle \multimap\!\!\!\!\!\!\leftrightarrow \langle s \cdot x \mid y' \rangle$ and we have that $s \perp x \otimes y'$. Thus, $s \in (P \otimes AY^\perp)^\perp$.

(2) $\Leftrightarrow$ (4). The equivalence holds by a similar argument, since $\langle s \mid x \otimes y' \rangle \multimap\!\!\!\!\!\!\leftrightarrow \langle s^\perp \mid y' \otimes x \rangle$.

(3) $\wedge$ (4) $\Rightarrow$ (1). Assume that s satisfies (3) and (4). First, we know from the proof of (3) $\Rightarrow$ (2) that $s \in (P \otimes AY^\perp)^\perp$. Second, by the implication (2) $\Rightarrow$ (4) applied on P and $AY^\perp$, we know that for all $y' \in AY^\perp$, $s^\perp \cdot y'$ is defined and belongs to $AX^\perp$. So by the characterization of Proposition 22, we conclude that $s \in A(X \multimap Y)$.

Bibliographic context

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